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What does the occluding contour tell us about solid shape?

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Abstract. A new theorem is discussed that relates the apparent curvature of the occluding contour of a visual shape to the intrinsic curvature of the surface and the radial curvature. This theorem allows the formulation of general laws for the apparent curvature, independent of viewing distance and regardless of the fact that the rim (the boundary between the visible and invisible parts of the object) is a general, thus twisted, space curve. Consequently convexities, concavities, or inflextions of contours in the retinal image allow the observer to draw inferences about local surface geometry with certainty. These results appear to be counterintuitive, witness to the treatment of the problem by recent authors. It is demonstrated how well-known examples, used to show how concavities and convexities of the contour have no obvious relation to solid shape, are actually good illustrations of the fact that convexities are due to local ovoid shapes, concavities to local saddle shapes.

1 Introduction

Three-dimensional Euclidean geometry is so complicated that many people experience difficulties in obtaining veridical notions about unfamiliar spatial relationships. Oddly enough, our visual systems automatically interpret some complicated spatial configurations correctly, despite the fact that we may hold primitive and even incorrect intellectual notions about geometrical relations. One such instance is the interpretation of the visual contours of smooth curved objects. Here, even scientific investigators appear as a rule to hold quite false and unfruitful notions. Despite this fact (which will be illustrated later) even these investigators have quite normal visual impressions; this is duly noted by them as a remarkable and as yet unexplained fact. Because the geometry is incompletely or wrongly understood, it is thought that the performance of the visual system must be due to (uncannily lucky) guesswork or the use of prior information and cannot be based on causal relationships.

In this paper I intend to discuss some common fallacies in this respect, and to replace them with factual observations. Some of the latter are not at all new, but apparently only known to mathematicians (differential geometers) and not to visual scientists. One result I will present is (as far as I could ascertain) new, and of recurring applicability in matters of visual perception of solid shape. In order to convince the reader of the need for a thorough analysis I need a few quotations, and I have selected them from the well-known book of the late David Marr (1982), which is so widely appreciated that I could hardly detract from its importance.

2 Preliminary

Consider smooth rigid bodies, for which the tangent plane is defined for every point of the surface and moreover is a smooth (as many times differentiable as necessary) function of position. From a given position in space all the rays through the vantage point that are tangent to the surface of the body can be constructed. These rays form a conical surface, the *contour* (or, equivalently, the curve through which this conical surface cuts the unit sphere around the vantage point may be regarded as the contour). The rays touch the object along a smooth curve on its surface, the *rim*.

Although the rim is smooth, the contour need not be: it can display cusps (Koenderink and van Doorn 1976, 1979, 1984); moreover, the contour may intersect itself, which the rim cannot do.

It is convenient to limit the discussion to *generic* situations. By this, I mean situations which are stable under arbitrarily small deformations, for example, small excursions of the vantage point or small deformations of the object. For instance, a line may have infinitely many points in common with a plane, but this is not a generic situation, for the smallest deformation of the plane will destroy it. The generic tangent rays are of three types: either they graze the surface and meet it at two infinitesimally close points (like a ray grazing a sphere); they graze the surface and enter it, meeting the surface at three infinitesimally close points [then, the contour cusps and the visible contour ends (Koenderink and van Doorn 1982)]; or, finally, they may graze the surface at two places which are separated by a finite distance (this leads to the so-called 'T-junctions'). Of the latter types of ray there exist only finitely many (for a fixed vantage point), of the first type there is a one-dimensional manifold: the cone mantle of the contour. The latter two types thus are isolated singular rays of the contour.

The rim is not a plane curve, it is also not everywhere perpendicular to the visual direction (for example, at the cusps of the contour the rim runs *along* the ray). It is thus a space curve and generically has an everywhere positive curvature and finitely many isolated points of vanishing torsion (points at which the 'helicity' changes from a right-handed to a left-handed screw sense, or vice versa). Thus it is strictly senseless to speak of inflections of the rim, although it would be a valid question to ask for its torsion zeros. The contour can be understood as a spherical curve, thus it makes good sense to speak of *its* inflections.

3 Inflections of the contour and inflections of the surface

Even in the absence of a metric or a coordinate system, it makes sense to speak of the convexities and concavities (and thus inflections) of the contour. Marr (1982) notes these features as the only ones that are straightforwardly available, rightly I think. However, he adds (p 221): "In general, of course, points of inflexion in a contour need have no significance for the surface". This is just not true. Marr illustrates his contention with a snake whose body varies in distance from the vantage point, then notes that convexities belong to near, concavities to far parts (see figure 3, upper part). Whereas this is a valid observation, the conclusion is wrong: the convex parts of the contour bound synclastic surface patches, the concave parts anticlastic surface patches (to be defined below). The confusion probably arises because the rim (called "contour generator" by Marr) moves over the body when you move the vantage point. Thus the simile with a bent wire that can be viewed from different positions is wrong because the space curve itself *changes* when the observer moves.

Consider the facts about the curvature of the contour. First, consider the case of a distant vantage point. Suppose one views a rim point R from a direction ξ. The following are constructed: the normal plane ν, defined by the normal to the surface at R and the visual direction ξ, and the plane of projection π perpendicular to ξ. Then two curves defined through these entities may be regarded: the curve λ in which the normal plane cuts the surface, and the curve μ along which the cylinder of tangent rays parallel to ξ cuts the plane of projection π (figure 1).

The radii of curvature of these curves at R or its projection may be measured. The curvature

$$K_r = 1/r_r,$$

where r_r is the radius of curvature of λ, is the *radial curvature* of the surface, the curvature along the line of sight;

$$K_t = 1/r_t,$$

where r_t is the radius of curvature of μ, is the *transverse curvature,* the curvature of the projection. The marvelously simple relation can then be derived (Appendix I):

$$K_r K_t = K,$$

where K denotes the *Gaussian curvature* of the surface. Before entering into a discussion on the definition and importance of K, I note another important relation (Appendix II). If the projection is from a finite distance d, the *apparent curvature* of the contour can be found from the simple relation:

$$K_{app} = dK_t = \frac{dK}{K_r}.$$

The meaning of K_{app} is that $\arctan(1/K_{app})$, expressed in radians, denotes the radius of the apparent curvature of the contour in the optic array. (The radius of apparent curvature of the contour is $\arctan(1/K_{app}) \times 180/\pi$ deg visual angle.) Thus, the horizon has a radius of 90 deg ($K_{app} = 0$), whereas the apparent curvature of the moon's disc is 230, which is merely the reciprocal of its radius (0.25 deg) in radians. It may be noted that the apparent curvature of the contour only vanishes at places where $K = 0$ in realistic cases ($K_r < \infty$; $d > 0$). Thus *inflections of the contour correspond to inflections ($K = 0$) of the surface.* This is generally true and a restriction to planar rims and effectively infinite viewing distances is not at all necessary.

The implications of the above rule are the following. Whenever $K > 0$ the surface is either a concavity (like the inside of an eggshell) or a convexity (like the outside of an eggshell). Whenever $K < 0$ the surface is saddle-shaped. For concavities and convexities the surface is curved in the same sense for all directions, hence the term *synclastic.* For a saddle shape, the surface is curved one way for some directions and the opposite way for other directions: hence the term *anticlastic.* For a saddle shape there exist two directions for which a cut through the surface does reveal zero curvature: these are the *asymptotic* directions. When the Gaussian curvature vanishes ($K = 0$) the surface is locally cylindrical (curved only in one direction), hence the term *monoclastic.* Generically, synclastic and anticlastic points occur in regions, their boundaries are the *flexional curves* on which the monoclastic points are situated. These flexional curves are the surface analogues of the inflections of plane curves (Hilbert and Cohn-Vossen 1932). They are of great importance in the

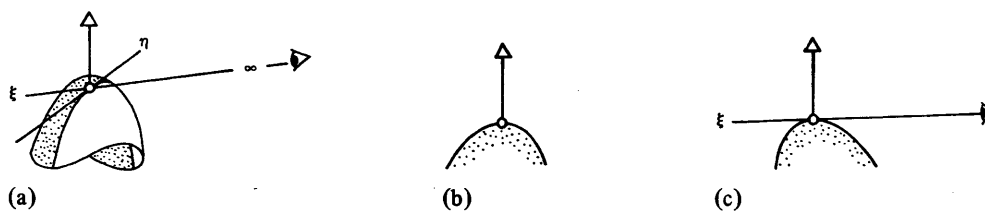


Figure 1. (a) A surface seen from a visual direction ξ, the eye at infinity. The shaded region cannot be seen from that vantage point, its border is the *rim.* A tangent ray at the rim is not (in general) perpendicular to the visual direction (η not perpendicular to ξ). (b) The projection of the rim on a plane π, normal to the visual direction. Its curvature is K_t. (c) The object cut with a plane ν defined by the visual direction ξ and the normal at the surface. Its curvature is K_r.

mathematical description of solid shape and are also recognized in artistic practice (Koenderink and van Doorn 1980, 1982, 1984). In the literature on differential geometry the reader will often find the terms elliptic, hyperbolic, and parabolic for synclastic, anticlastic, and monoclastic regions, respectively. I prefer the latter, more descriptive, terms which also have been in continued use in the mathematical literature on the study of surfaces in three-dimensional space.

4 The general rules of contour curvature

Consider the consequences of the general formula

$$K_{\text{app}} = \frac{dK}{K_{\text{r}}} .$$

First note that the radial curvature K_{r} cannot change sign at a contour, because the object occludes the sight of farther parts (figure 2). Let us treat this curvature as positive (an arbitrary choice). The distance d is also a positive number, thus the sign of K_{app} is the same as the sign of the Gaussian curvature.

First consider the case $K > 0$: concavities and convexities. For a concavity the case $K_{\text{r}} > 0$ can never occur, thus concavities can never occur at the contour. For convexities $K_{\text{app}} > 0$ necessarily; let us use this to define *convexity* of the contour (figure 2). For the case $K < 0$ there exists only a certain set of visual directions for which $K_{\text{r}} > 0$. For such cases $K_{\text{app}} < 0$ necessarily; thus the contour displays a *concavity* (figure 2).

As a result, the general rule can be stated: *convexities of the contour correspond to convex surface patches, concavities to saddle-shaped surface patches.* There are no exceptions to this rule; it is universally valid for all vantage points and irrespective of the fact that the rim is, in general, a general space curve. Note that the requirements of a planar rim and an effectively infinite viewing distance formulated by Marr as necessary conditions for an interpretation of apparent curvature are in fact quite unnecessary. This is a happy circumstance because such conditions would limit the cases that can be treated at all to such a small set as to be without practical use. That the requirement of infinite viewing distance is actually unnecessary can be understood without any computation at all: theorems about the sign of the curvature of the contour are theorems about the contact of the surface with planes and lines, thus invariant against projective transformations. Any theorem for distant vantage points can easily be transformed to a theorem about a close vantage point and vice versa.

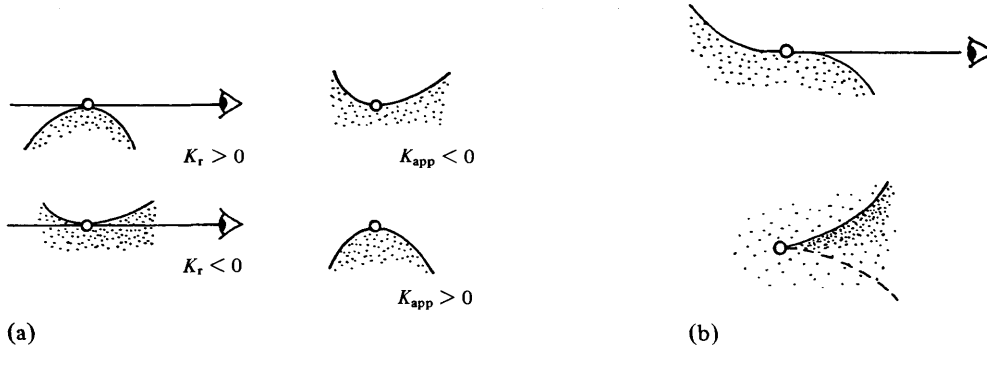


Figure 2. (a) Conventions of the sign of curvatures used in the text. Left: only a positive radial curvature leads to a visual contour. Right: a negative apparent curvature is a concavity, a positive apparent curvature a convexity. (b) Top: the visual direction along an asymptotic ray. The normal cut is shown, the radial curvature vanishes. The ray touches the surface, yet enters it. Bottom: the same case in projection: the contour is cusped (infinite apparent curvature), and an *ending contour* is obtained.

Marr's example of the 'snake' may be modelled with a torus, cut into two and pasted together again after a transposition (figure 3). For the near part the rim runs through the synclastic region, for the far part in the anticlastic region. Thus convexities and concavities of the contour neatly correspond to synclastic and anticlastic patches of the snake: there is a simple relation between shape of contour and shape of surface. That artists are sensitive to the transition from one kind of curvature to the other may be seen from the way the hatching changes in the tail of the lion in Dürer's "Samson killing the lion" (1498) as shown in figure 4.

The case $K < 0$ permits another possibility that is of great interest: there exist two directions for which K_r vanishes: the asymptotic directions. For such a direction the ray is tangent to the surface, yet it *enters* the surface (figure 2b). Our general formula yields $K_{app} = \infty$: an infinite curvature. The contour has a cusp at such a point

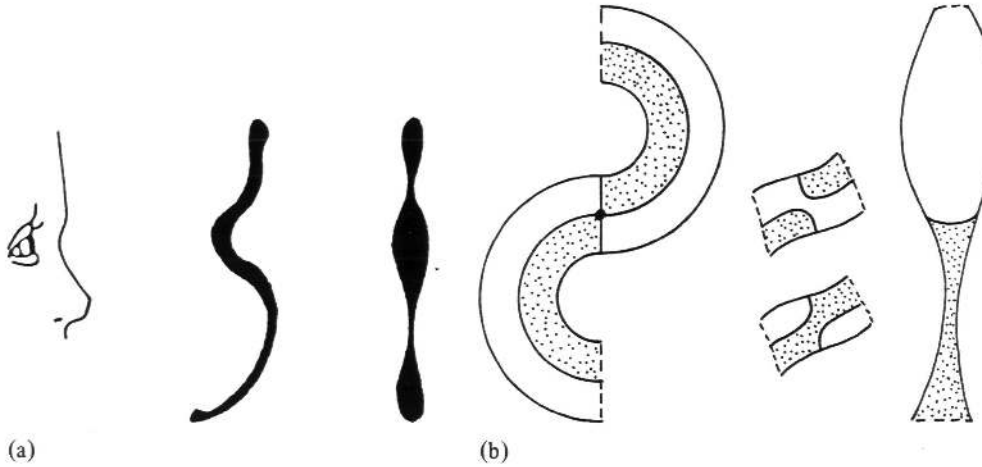


Figure 3. (a) A figure taken from Marr (1982). The suggestion is that convexities and concavities in the projection of the snake have to do with relative *distances* rather than with local shapes. (b) A torus cut into two and pasted together again. The shaded regions are anticlastic, the other regions synclastic. The small insets show the generic case after a small deformation. In projection (right), this 'snake' has convexities where the body is locally egg-shaped, concavities where the body is locally saddle-shaped, inflexions at flexional curves of the body.

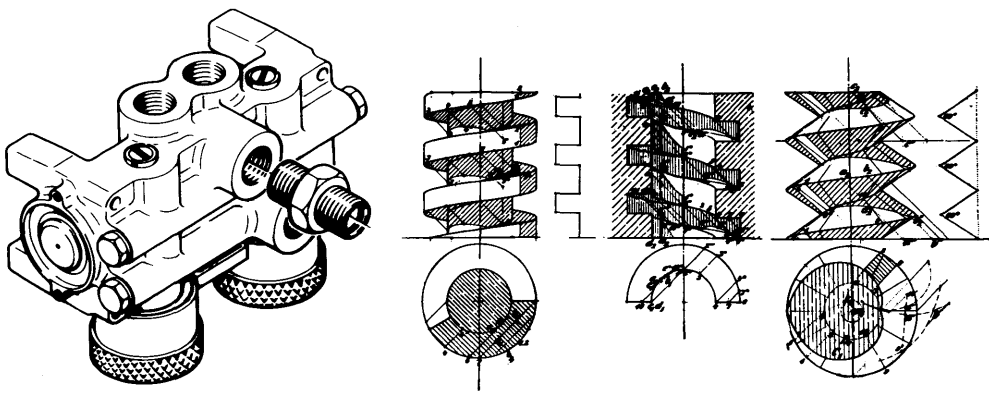
Figure 4. Details from Dürer's "Samson killing the lion". (Bartsch #2; the print dates from 1498.)

and only one branch of the cusp is visible (figure 2b). Thus this case corresponds to an ending contour. The rim is quite regular at such points, without any cusp. The cusp in the contour is a result of the fact that the rim runs along the ray: this is possible because the rim is a space curve. For objects with only planar curves as possible rims (such as a sphere) an ending contour can never be obtained.

5 Implications of planarity and nonplanarity of the rim

Incorrect assumptions about geometrical configurations leading to convexities and concavities of the contour have led several writers to the notion that an assumption of planarity of the rim is necessary if one is to make sense of these flexuosities. For example, Marr (1982, p 222) states: "This third assumption is a strong one that sharply delimits the class of surfaces whose shape can be interpreted by silhouette. However, it seems unavoidable if we wish to distinguish convex and concave segments in the interpretation process". I shall discuss the restrictions created by such an assumption here, but first of all deal with some further speculations related to this point. It has been remarked that the planar condition is actually embodied in most modern design. This is untrue if you include such common shapes as cuboids (the rim for a cube is only planar for a restricted set of vantage points). It has also been remarked that all of the outlines drawn in mechanical engineering diagrams satisfy this condition. In fact the perusal of any book on engineering drawing will reveal many exceptions (for example figure 5). Moreover, it is held that if the condition of rim planarity is violated, we do seem to get the shape wrong. As an example it is remarked that a hexagon is not spontaneously perceived as the projection of a cube. This is certainly true, but it is also true that those projections of a cube for which the rim is planar are seldom perceived as projections of a cube, whereas 'generic' projections which show little symmetry and have nonplanar rims are the most easily interpreted as projections of cubes. For example, figure 6 shows an illustration from a technical book on drawing in which the artist is advised to choose generic projections—with nonplanar rims—for an optimum suggestion of solid shape.

A set of objects which display planar rims for a restricted set of vantage points are the so-called 'generalized cones', introduced by Binford (1971) and employed by many authors since. The torus is an example of such an object. It is often bluntly



(a) (b)

Figure 5. Examples from technical drawing. These are just some random selections; numerous examples of nonplanar rims can be found in any book on technical drawing methods. (a) Figure from the leaflet on Rotring drawing implements. Isometric after DIN 5. (b) Figure from "Audels Mechanical Drawing Guide" by Audel Staff, Theo. Audel and Cp, Publ., New York, 1947, p 88.)

stated that a generalized cone automatically produces planar rims. This is not the case; however, an added restriction is that the axis is not foreshortened. For a general generalized cone, such as the torus, this meant that the rim is planar only for a single direction of view. This is obvious for a typical projection of the torus which shows two ending contours (figure 7): at these points the rim runs in the visual direction, thus if the rim were indeed planar the eye would be in the plane of that curve and the contour would degenerate to a segment of a great circle of the optic array which is apparently not the case.

Thus the (unnecessary) restriction to planar rims not only restricts the choice of objects to generalized cones, it also severely restricts the permitted directions of view (often to a single one). Thus, *almost all visual directions are forbidden.* "Stated baldly, this result means that, in general, shape cannot be derived from occluding contours alone unless that shape is made from generalized cones and is viewed from a position from where its axis is not foreshortened" (Marr 1982, p 223). This appears

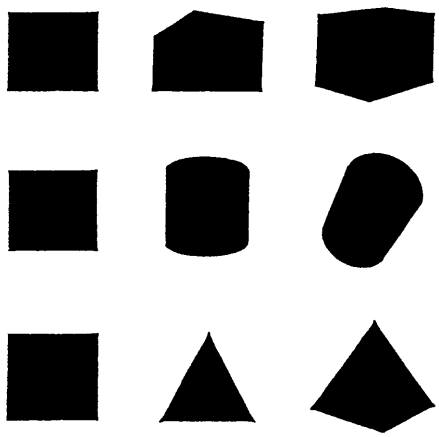


Figure 6. An instructive figure taken from a book on drawing (Hogarth 1981, p 11). Note how the strongest spatial impressions arise from the 'generic' views for which the rim is everything but a plane curve. The original legend runs:
"*Geometrical Silhouettes*
Some silhouettes are ambiguous and some are not. In the first column at the left, the three squares are ambiguous because they do not suggest specific three-dimensional forms. It is not clear whether they represent cubes, a head-on view of a cylinder, or the rectangular base of a pyramid. The middle column represents a slight shift in viewpoint, with less visual confusion: the three silhouettes do, in fact, suggest a cubic form, a cylinder, and a pyramid. But the three silhouettes in the righthand column are clearer still, particularly in the case of the pyramid, which now obviously rests on a square base".

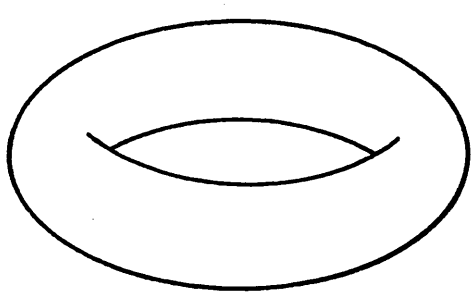


Figure 7. A typical view of a torus. Note the two ending contours. At these points the rim is along the visual direction. The rim is a twisted space curve.

to be a meagre basis for the claim that "... a natural link exists between generalized cones and the imaging process itself" and that "generalized cones will play an intimate role in the development of vision theory" (Marr 1982, p 223).

It may be that Marr was really looking for minimal constraints that would permit the computation of global three-dimensional shape from a single projection. This problem is clearly highly degenerate and any solution would depend more on the arbitrary constraints (for example, that the shape be a generalized cylinder) than on the actual data. In contradistinction to the global problem, local shape (that is, the three-dimensional shape of regions on the rim) can be inferred from the data in a consistent way.

The facts are that the only shapes which yield planar rims for *any* vantage point are the developable surfaces (planes, cylinders, cones, tangent surfaces of space curves) and the quadrics (the ellipsoids, paraboloids, and hyperboloids). This has been proved by Blaschke (1923).

6 Conclusions

In this paper I have discussed the geometrical facts about the implications of the concavities and convexities of visual contours for the inference of the local shape of surface regions. I have done this with the help of a simple but powerful theorem about surface curvatures that appears to be novel. It is intimately related to two results of differential geometry: one well-known and mentioned in any elementary treatise, namely Euler's theorem; the other rather obscure and mentioned in only a few texts, namely Blaschke's "dual" of Euler's theorem. (The theorem $K_t K_r = K$ does not appear in any text I know of, however.) The result is very simple and clear-cut: for *any* vantage point and *without any restriction* on the shape of the rim, a convexity of the contour corresponds to a convex patch of the surface, and a concavity to a saddle-shaped patch. Inflections of the contour correspond to the flexional curves of the surface. Thus the flexuosities of the contour signify actual properties of the surface, a fact that does not seem to have been appreciated by workers in the field of vision before: although I have quoted some remarks from Marr's (1982) text, I could have used many other sources.

The theorem about the relation between the radial and the projected curvatures is of interest in another way, not noted above: because the Gaussian curvature is invariant against arbitrary bending of the surface, the product of these curvatures is also invariant against bending. Thus the theorem can be extended to a class of nonrigid shapes. Here, 'bending' means any transformation that conserves the metric as measured *along the surface*. Thus paper and flexible steel shells can easily be subjected to bendings, but with rubber membranes care must be taken to avoid *stretching*. As a result one has the theorem that arbitrary bendings cannot change convexities of the contour into concavities and vice versa. (This can easily be verified if one cuts a cheap rubber football into two. Through bending one can easily change the radial and transverse curvature over a certain range, but one cannot produce a concave contour without stretching.) The apparent curvature of the contour really enables one to infer intrinsic properties of the surface geometry.

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Appendix

1 Projected and radial curvature

Consider first the case of a distant (effectively infinitely distant) vantage point, that is, parallel projection. Let the direction of sight be the x direction. Consider a point P on the rim, and the z direction in the direction of the normal. Then the x-y plane is the tangent plane at P. Let P be the origin of the Cartesian coordinate system, then one has as description of the surface near P:

$$z(x, y) = \tfrac{1}{2}(ax^2 + 2bxy + cy^2) + (\text{third-order and higher terms}). \tag{A1}$$

The locus of the *rim* is defined through the condition

$$\frac{\partial z}{\partial x} = ax + by + (\text{second-order and higher terms}) = 0, \tag{A2}$$

because the x direction grazes the surface at the rim. Note that the rim runs in the direction that makes an angle $\varphi = \arctan(-a/b)$ with the positive x direction. Thus the rim is not, in general, perpendicular to the line of sight; one says it is "conjugated" to that direction.

When x is eliminated from equations (A1) and (A2), one obtains

$$z = \tfrac{1}{2}\frac{ac - b^2}{a} y^2 = \tfrac{1}{2}K_t y^2, \tag{A3}$$

(neglecting third-order and higher terms), which is the analytical expression that describes the projection of the rim on the y-z plane, in other words, the *contour*. The curvature of the projection is thus

$$K_t = \frac{ac - b^2}{a}. \tag{A4}$$

The normal plane through the line of sight is the x-z plane. The surface meets this plane in the curve

$$z = \tfrac{1}{2}ax^2 = \tfrac{1}{2}K_r x^2, \tag{A5}$$

if third-order and higher terms are neglected. Thus the radial curvature is

$$K_r = a. \tag{A6}$$

From equations (A4) and (A6) one obtains immediately the important result

$$K_r K_t = ac - b^2 = \frac{(\partial^2 z/\partial x^2)(\partial^2 z/\partial y^2) - (\partial^2 z/\partial x \partial y)^2}{1 + (\partial z/\partial x)^2 + (\partial z/\partial y)^2} = K, \tag{A7}$$

as $\partial z/\partial x = \partial z/\partial y = 0$ in P and $\partial^2 z/\partial x^2 = a$, $\partial^2 z/\partial y^2 = c$, and $\partial^2 z/\partial x \partial y = b$.

2 The apparent curvature

Let ξ be a tangent ray through P, and let Q be any point on ξ. Let the distance PQ be d. From Q other tangent rays may be drawn; these rays are the generators of a cone with vertex Q that touches the surface along the rim. Consider the second-order osculating cone, the second-order cone that approximates the real one most closely. When one lets Q range over ξ, the semitop angle of this osculating cone varies. This always occurs in a neat, predictable manner, however, A useful theorem, originally due to Mannheim but first well described by Blaschke (1916), states: *all osculating cones envelop a sphere.* This sphere may be called the 'Mannheim sphere'.

When Q is at infinity ($d \to \infty$) the cone degenerates into a cylinder with radius K_t^{-1}. This must also be the radius of the Mannheim sphere, for a cylinder touches a sphere along a great circle.

Consider figure A1. The Mannheim sphere at P is drawn, with radius $R_t = K_t^{-1}$. The sphere obviously touches the tangent plane, thus the normal to the sphere is the normal to the surface (at P). Also drawn is an osculating cone for a point Q at distance d. From simple geometry one obtains the semitop angle, α:

$$\alpha = \arctan\left(\frac{1}{dK_t}\right) .$$

Around Q is constructed a unit sphere which represents the manifold of visual directions for the point Q. The osculating cone cuts this unit sphere in a small circle. The (spherical) curvature of this small circle is the *apparent curvature* of the contour.

To calculate the apparent curvature one must find the geodesic curvature of this small circle. It can be shown (for example, Lipschutz 1969) to be given by

$$K_{\text{app}} = \cot \alpha.$$

Consequently, one obtains

$$K_{\text{app}} = dK_t,$$

that is, the formula used in the text. (Thus the radius of apparent curvature is $K_{\text{app}}^{-1} = (dK_t)^{-1}$ radians, or $180/\pi dK_t$ deg visual angle.)

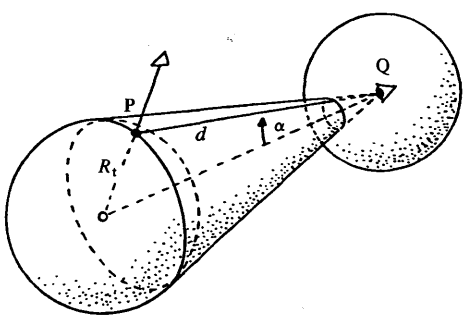

Figure A1.

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